Վարժարան

VARZHARAN · ONLINE ACADEMY OF MATHEMATICS

COURSE SYLLABUS
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Statistics

Վիճակագրություն

COURSE CODE	MTH-ST
PROGRAM	High School Mathematics · Ավագ դպրոցի մաթեմատիկա
LEVEL	Standalone course · typically Grades 10–12
DURATION	90–108 instructional hours (approx. 78 lessons)
PREREQUISITE	Algebra I (MTH-A1); Algebra II recommended but not required.
LEADS TO	AP Statistics (MTH-APS) for students seeking college credit

COURSE DESCRIPTION

Դասընթացի նկարագրություն

This is a full introductory statistics course for students who want quantitative reasoning without the pace and examination pressure of the AP course, or who intend to take AP Statistics later and want a solid first pass. It covers exploratory data analysis, study design, probability, random variables and sampling distributions, confidence intervals, hypothesis testing, and inference for means, proportions, and relationships. The emphasis throughout is on interpretation and on knowing what a conclusion is and is not entitled to claim.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Statistics is the branch of mathematics students are most likely to use and most likely to be misled by. Վարժարան teaches it as a discipline of careful claims: every technique is introduced with the question it answers, every conclusion is stated in context with its uncertainty attached, and every result is examined for what it cannot establish. Computation is deliberately secondary — the calculator or software performs it — and the assessed skill is choosing the right procedure, checking that its conditions are met, and stating the conclusion honestly. A student who can compute a p-value but cannot say what it means has learned nothing of value.

PLACEMENT AND READINESS INDICATORS

Ընդունելություն և պատրաստվածություն

The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Computes with fractions, decimals, and percents fluently	Probability and proportion work depend on it continuously.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Reads and constructs graphs and interprets slope	Prerequisite for regression in Unit 3.
Solves linear equations and works with formulas	Sufficient algebra for the whole course; heavy manipulation is not required.
Reads a quantitative passage carefully and identifies the claim	The most predictive readiness indicator for this course, and the least often assessed.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

1. Distinguish categorical from quantitative variables and choose appropriate displays for each.
2. Describe a distribution by shape, centre, spread, and unusual features, in context.
3. Compute and interpret measures of centre and spread and assess the effect of outliers.
4. Use the normal distribution and z-scores to compute proportions and percentiles.
5. Analyze bivariate data with scatter plots, correlation, and least-squares regression.
6. Interpret regression output, examine residuals, and recognize influential points.
7. Distinguish observational studies from experiments and identify what each can establish.
8. Design a study with appropriate randomization, control, replication, and blocking.
9. Apply the rules of probability, including conditional probability and independence.
10. Work with discrete random variables, expected value, and the binomial and geometric distributions.
11. Understand sampling distributions and the Central Limit Theorem.
12. Construct and interpret confidence intervals for means and proportions.
13. Carry out significance tests for means and proportions and interpret p-values correctly.
14. Perform chi-square tests for categorical data and inference for the slope of a regression line.
15. Critique a statistical claim made in a report or article and identify its limitations.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 12-13 LESSONS

Exploring Data

Տվյալների ուսումնասիրություն

TOPICS COVERED

- Individuals, variables, categorical and quantitative data
- Bar charts, pie charts, and two-way tables for categorical data
- Marginal and conditional distributions; segmented bar charts
- Dot plots, stem plots, and histograms for quantitative data
- Choosing an appropriate display and bin width
- Describing a distribution: shape, centre, spread, outliers
- Mean, median, and the effect of skew
- Range, interquartile range, standard deviation, and variance
- The 1.5 × IQR rule for identifying outliers
- Five-number summaries and box plots; comparing distributions
- Resistant and non-resistant measures
- The effect of linear transformations on centre and spread
- Density curves and the normal distribution
- The empirical rule and z-scores
- Normal probability calculations and inverse normal
- Assessing normality with a normal probability plot

LEARNING OBJECTIVES

1. Choose and construct an appropriate display for any variable type and justify the choice.
2. Describe a distribution completely in context, naming shape, centre, spread, and unusual features.
3. Compute and interpret a z-score and use the normal distribution to find proportions and percentiles.
4. Explain why the median and IQR are preferred for skewed data.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces a complete contextual description of a distribution in 9 of 10 tasks.
- Performs normal distribution calculations at 90% accuracy in both directions.
- Chooses resistant or non-resistant measures appropriately with justification in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A distribution is described by centre alone, with no mention of spread or shape.	Require all four elements in every description, in context. Make it a written checklist until it is automatic.

MISCONCEPTION	RECOMMENDED RESPONSE
The normal distribution is assumed for any unimodal data.	Check with a normal probability plot. Many real distributions are unimodal and clearly non-normal.
Standard deviation is interpreted as the average distance from the mean.	It is close to that but computed with squares. The distinction matters when explaining why it is non-resistant.

TEACHING NOTE

Descriptions in context are the assessed skill, not the arithmetic. A numerically correct answer with no contextual sentence should receive partial credit at most — this standard, set in Unit 1, shapes the entire course.

UNIT 2 · 9-10 LESSONS

Sampling and Experimental Design

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TOPICS COVERED

- Populations, samples, parameters, and statistics
- Census versus sample
- Simple random samples and how to select one
- Stratified, cluster, systematic, and multistage sampling
- Convenience and voluntary response samples and why they fail
- Undercoverage, nonresponse, and response bias
- Question wording effects
- Observational studies versus experiments
- Explanatory and response variables; confounding
- The principles of experimental design: control, randomization, replication
- Completely randomized designs
- Blocking and matched pairs designs
- Blinding, double blinding, and the placebo effect
- Scope of inference: to what population, and can causation be claimed
- Ethical considerations in study design

LEARNING OBJECTIVES

1. Identify the sampling method used in a described study and name any resulting bias.
2. Design a completely randomized experiment for a stated question, specifying all three principles.
3. Explain when blocking improves a design and construct a blocked design.
4. State the scope of inference for a given study, including whether causation may be claimed.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Identifies sampling methods and bias correctly in 9 of 10 described studies.
- Designs a valid experiment including all three principles in 4 of 5 tasks.
- States the scope of inference correctly in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Random assignment and random selection are treated as the same thing.	Random selection permits generalization to a population; random assignment permits causal claims. They are different and independently necessary.
Any large sample is taken to be representative.	Bias is a property of the method, not of the size. A biased method with a large sample is confidently wrong.
Blocking is confused with stratification.	Stratification is a sampling technique; blocking is an experimental design technique. The purpose is analogous but the setting differs.

> **TEACHING NOTE**
> *This unit contains no computation and is often undervalued by students for that reason. It is the unit that determines whether any subsequent conclusion is worth anything at all, and it should be assessed as rigorously as the computational units.*

UNIT 3 · 10-11 LESSONS

Bivariate Data and Regression

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TOPICS COVERED

- Scatter plots and describing form, direction, strength, and outliers
- The correlation coefficient r and its properties
- What correlation does not measure
- Least-squares regression and the meaning of 'least squares'
- Interpreting slope and intercept in context
- Prediction and the dangers of extrapolation
- Residuals and residual plots
- Using residual plots to assess model appropriateness
- The coefficient of determination r^2 and its interpretation
- Outliers, high-leverage points, and influential observations
- Transforming data to achieve linearity: logarithmic and power transformations
- Choosing among competing models
- Correlation and causation; lurking and confounding variables

LEARNING OBJECTIVES

1. Interpret r, r^2, slope, and intercept correctly in context.
2. Use a residual plot to decide whether a linear model is appropriate.
3. Distinguish an outlier from a high-leverage point from an influential observation.
4. Transform data to linearize a relationship and interpret the transformed model.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Interprets all regression output components in context in 9 of 10 trials.
- Assesses model appropriateness from residual plots in 4 of 5 tasks.

- Transforms and interprets nonlinear relationships correctly in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
r^2 is interpreted as the percentage of points on the line.	It is the proportion of variability in the response explained by the model. The wording must be precise and should be practiced.
A high r is taken as confirmation that a linear model is appropriate.	The residual plot decides that, not r. Curved data can produce a high correlation.
Causation is inferred from a strong regression relationship in observational data.	Only a randomized experiment supports causation. Require a plausible confounder to be named for every observational claim.

TEACHING NOTE

Requiring the student to name a plausible confounding variable for every observational claim is the most transferable habit in the entire course. It is also the one most likely to be used weekly for the rest of their life.

UNIT 4 · 12-13 LESSONS

Probability and Random Variables

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TOPICS COVERED

- Randomness, probability, and the law of large numbers
- Simulation as a tool for estimating probability
- Sample spaces, events, and basic probability rules
- Complement, union, and intersection; the general addition rule
- Conditional probability and the general multiplication rule
- Independence and how to test for it
- Tree diagrams and two-way tables
- Bayes-type reasoning in accessible contexts
- Discrete random variables and probability distributions
- Expected value and standard deviation of a random variable
- Linear transformations and combinations of random variables
- The binomial setting, distribution, mean, and standard deviation
- The geometric setting and distribution
- Using the normal approximation where appropriate
- Continuous random variables and density curves

LEARNING OBJECTIVES

1. Compute conditional probabilities and test independence formally.
2. Design and carry out a simulation to estimate a probability.
3. Compute the expected value and standard deviation of a discrete random variable and interpret both.
4. Recognize a binomial or geometric setting and compute the associated probabilities.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes conditional probabilities and tests independence at 90% accuracy.
- Recognizes binomial and geometric settings correctly in 9 of 10 trials.
- Computes and interprets expected value in context in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Independent and mutually exclusive are treated as synonyms.	Mutually exclusive events with non-zero probability are dependent — knowing one occurred tells you the other did not. Construct both examples.
The binomial conditions are not checked before the formula is used.	Fixed number of trials, two outcomes, constant probability, independence. Require all four to be verified in writing.
Expected value is interpreted as the value most likely to occur.	It is a long-run average and may be a value that can never occur. Use a lottery example.

TEACHING NOTE

Simulation should be used throughout this unit rather than treated as a separate topic. Students who can simulate a situation understand its structure, and simulation also builds the intuition needed for sampling distributions in Unit 5.

UNIT 5 · 12-13 LESSONS

Sampling Distributions and Confidence Intervals

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TOPICS COVERED

- Parameters and statistics revisited
- Sampling variability and the idea of a sampling distribution
- Simulating a sampling distribution
- Bias and variability of an estimator
- The sampling distribution of a sample proportion and its conditions
- The sampling distribution of a sample mean
- The Central Limit Theorem and what it does and does not say
- The logic of a confidence interval
- Confidence intervals for a proportion
- The margin of error and what affects it
- Choosing a sample size for a desired margin of error
- The t-distribution and why it is needed
- Confidence intervals for a mean
- Confidence intervals for a difference of two proportions
- Confidence intervals for a difference of two means
- Checking conditions before every interval
- Interpreting a confidence interval and a confidence level correctly

LEARNING OBJECTIVES

1. Explain what a sampling distribution is and describe its shape, centre, and spread.
2. State the Central Limit Theorem precisely and identify when it applies.
3. Construct a confidence interval after verifying all conditions.
4. Interpret both the interval and the confidence level in context, without overstating.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Verifies conditions correctly before every interval in 9 of 10 tasks.
- Constructs intervals for means, proportions, and differences at 90% accuracy.
- Produces a correct interpretation of both interval and confidence level in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A 95% confidence interval is described as containing 95% of the data.	It is a statement about the procedure's long-run capture rate for the parameter. Simulate 100 intervals and count how many capture the true value.
The Central Limit Theorem is applied to the population rather than to the sampling distribution.	It describes the distribution of the sample mean, not the data. Simulate with a strongly skewed population to make the distinction vivid.

MISCONCEPTION	RECOMMENDED RESPONSE
Conditions are omitted because the calculation works anyway.	The calculation always produces a number; the conditions determine whether the number means anything. Require the checks in writing, always.

TEACHING NOTE

The simulated sampling distribution is the single most valuable teaching device in the course. Building one — by hand at first, then by software — converts inference from a set of formulas into an idea the student can see.

UNIT 6 · 13-14 LESSONS

Significance Testing and Inference

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TOPICS COVERED

- The logic of a significance test
- Null and alternative hypotheses stated in context
- Test statistics and p-values
- Interpreting a p-value correctly
- Significance levels and the decision rule
- One-sample z-test for a proportion
- One-sample t-test for a mean
- Two-sample tests for proportions and means
- Paired data and the matched pairs t-test
- Type I and Type II errors and their consequences in context
- Power and the factors that affect it
- The relationship between confidence intervals and two-sided tests
- Chi-square goodness of fit test
- Chi-square tests for homogeneity and independence
- Inference for the slope of a least-squares regression line
- Checking conditions for every procedure
- Statistical significance versus practical importance
- Multiple testing and the danger of selective reporting
- Critiquing statistical claims in published reports

LEARNING OBJECTIVES

1. State hypotheses in context and carry out an appropriate test after checking conditions.
2. Interpret a p-value correctly, avoiding the common misstatements.
3. Describe Type I and Type II errors in the context of a specific study and state the consequence of each.
4. Critique a published statistical claim, identifying design, analysis, and interpretation weaknesses.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Selects and carries out the correct procedure with conditions checked in 9 of 10 tasks.
- Interprets p-values correctly in 9 of 10 trials.

- Produces a substantive critique of a published claim in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The p-value is interpreted as the probability that the null hypothesis is true.	It is the probability of data at least this extreme *if* the null were true. Rehearse the correct sentence until it is automatic; this is the most consequential misinterpretation in all of applied statistics.
Failing to reject the null is reported as proving the null true.	Absence of evidence is not evidence of absence. Require the phrase 'we do not have sufficient evidence' rather than 'there is no difference'.
Statistical significance is equated with practical importance.	A large enough sample makes trivial differences significant. Always report the effect size alongside the p-value.

> **TEACHING NOTE**
> *The correct interpretation of a p-value should be rehearsed aloud, in context, in every lesson of this unit. It is misstated constantly in professional publications; a student who states it correctly has acquired something genuinely rare.*

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Arithmetic fluency, graph reading, and quantitative reading comprehension.
Unit Check	End of each unit · 55 min	Unit mastery criteria; every item requires a contextual interpretation, not only a computation.
Data Analysis Project	Once per semester	The student poses a question, collects or obtains data, analyzes it, and writes a report with stated limitations.
Article Critique	Every third lesson	A short written critique of a real statistical claim in the press; the running record of the student's reasoning growth.
Mid-Course Assessment	After Unit 3	Cumulative Units 1–3.
End-of-Course Assessment	After Unit 6	All course outcomes, with an extended free-response section modelled on AP-style investigative tasks.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS
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TRACK	CADENCE	EXPECTED COMPLETION
Standard Year	2 lessons/week, 60 min	Full course in approximately 40 weeks.
Semester Intensive	3–4 lessons/week	Full course in approximately 20 weeks.
AP Preparation Track	2 lessons/week	Full course followed by the AP Statistics course (MTH-APS) with substantial overlap and faster pacing.
Support / Co-Seated	1–2 lessons/week alongside school	Paced to the student's school syllabus.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE
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Forty-five to sixty minutes per lesson. Every numerical answer must be accompanied by a sentence interpreting it in the context of the data; a bare number earns no credit. Every inference procedure must show the conditions checked before the computation. Assignments include one short article critique roughly every third lesson — a real claim from the press, analyzed for design, analysis, and interpretation weaknesses. This is the assignment students report as the most useful long after the course ends.

PROGRESS MILESTONES REPORTED TO PARENTS
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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- Describes a distribution completely and in context.
- Uses the normal distribution and z-scores in both directions.
- Identifies sampling methods and their biases.
- Designs experiments with control, randomization, and replication.
- States the scope of inference correctly for any study design.
- Interprets correlation, regression output, and residual plots correctly.
- Distinguishes correlation from causation and names plausible confounders.
- Applies probability rules including conditional probability and independence.
- Works with random variables, expected value, and binomial and geometric distributions.
- Explains sampling distributions and the Central Limit Theorem.
- Constructs and interprets confidence intervals with conditions verified.
- Carries out significance tests and interprets p-values correctly.
- Performs chi-square tests and inference for regression slope.
- Critiques published statistical claims substantively.

- Completed Statistics — ready for AP Statistics or college-level coursework.

MATERIALS AND RESOURCES

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- Graphing calculator with statistical capability (TI-84 family) or statistical software.
- Simulation tool for building sampling distributions — used extensively in Units 4 and 5.
- A curated collection of real data sets across science, economics, sport, and public health.
- A running file of press articles making statistical claims, for the critique assignments.
- Վարժարան Statistics problem bank — approximately 1,600 items, over half requiring written interpretation.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.